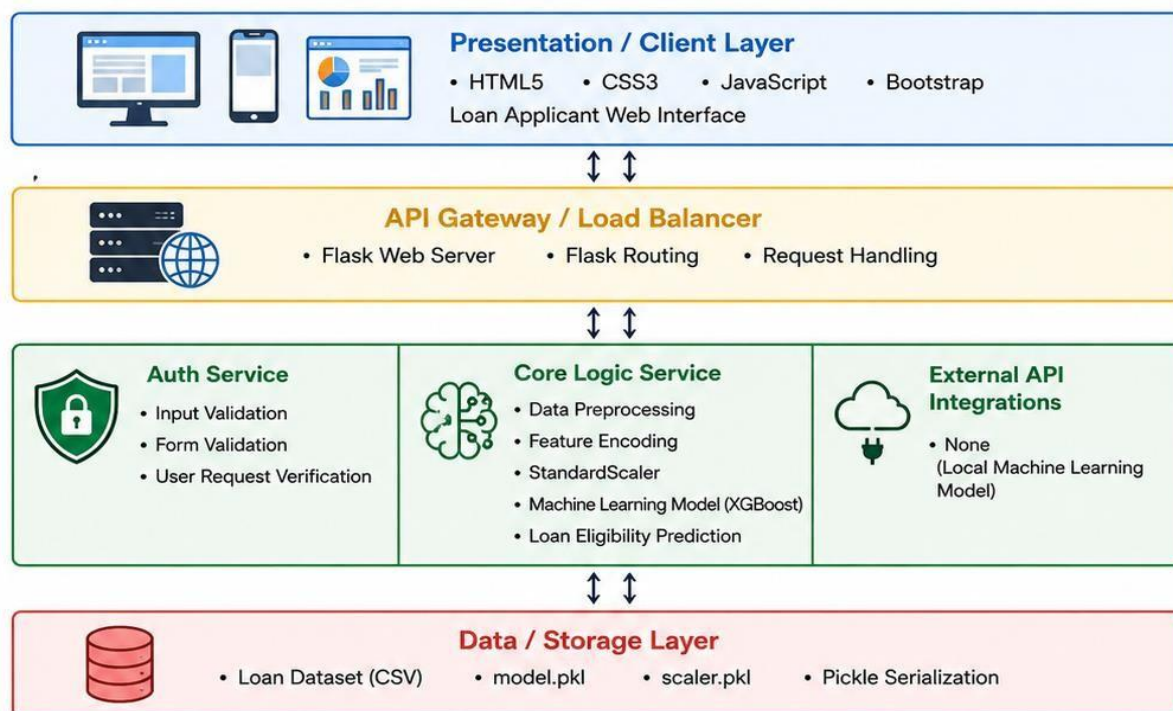


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a web interface where users enter loan details and view loan eligibility prediction results.	HTML5, CSS3, JavaScript, Bootstrap
API Gateway	Handles HTTP requests, routes user requests, validates input, and connects the frontend with the prediction engine.	Flask
Microservices (Core Logic Service)	Performs data preprocessing, feature encoding, scaling, loads the trained XGBoost model, predicts loan eligibility, and returns the result.	Python, Pandas, NumPy, Scikit-learn, XGBoost
Database / Data Storage	Stores the loan dataset, trained machine learning model, and scaler used for prediction.	CSV Dataset, model.pkl, scaler.pkl